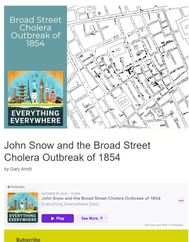
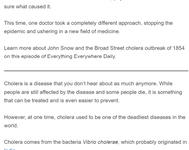


John Snow and the Broad Street Cholera Outbreak of 1854

by Gary Arndt



Podcast Transcript

In 1854 an unusually severe outbreak of cholera occurred in London.

While cholera was not an uncommon disease, physicians at the time weren't sure what caused it.

This time, one doctor took a completely different approach, stopping the epidemic and ushering in a new field of medicine.

Learn more about John Snow and the Broad Street cholera outbreak of 1854 on this episode of Everything Everywhere Daily.

Cholera is a disease that you don't hear about as much anymore. While people are still affected by the disease and some people die, it is something that can be treated and is even easier to prevent.

However, at one time, cholera used to be one of the deadliest diseases in the world.

Cholera comes from the bacteria *Vibrio cholerae*, which probably originated in India.

The bacteria causes an infection of the small intestine, resulting in severe diarrhea, vomiting, and muscle cramps. Cholera induced diarrhea can be so bad that it can result in severe dehydration and electrolyte imbalance, and it is possible in severe cases to go from first symptoms to death in a matter of hours.

Today, the fatality rates from cholera are low, but in the past, before treatments were developed, it was possible to see fatality rates as high as 50%.

So, just to lay the foundation for the story, cholera was a very nasty disease.

In the mid-19th century, doctors were very familiar with cholera but weren't sure what caused it.

Modern medicine hadn't yet been fully developed, and there were competing theories as to what caused cholera.

The first theory, and one which had been around for over 2,000 years, was the miasma theory.

The miasma theory held that diseases such as cholera and the plague were spread by foul air. The air was contaminated by rotting organic matter. Hence, if you were around a dead body or raw sewage, the miasmas which emanated from the organic materials were the cause of the disease.

The miasma theory dates back to at least ancient Greece in the 4th century BC and the Greek physician Hippocrates. The ancient Chinese also had a miasma theory, as did ancient India.

If you have ever seen plague masks that doctors used during the black death, which had enormous protrusions which looked like a bird's beak, they actually served a purpose. They were there to keep the miasma away.

So the miasma theory was well established and was the predominant theory among physicians at the time.

The other theory, which was relatively new, was the germ theory. The theory of germs was based on the discoveries of the Dutch scientist Antonie van Leeuwenhoek, who was the first person to discover single-cell organisms under a microscope.

The germ theory held that cholera was spread by these unseen microbes, which found their way from person to person via the water supply.

No one knew exactly what microbe was the culprit, but the germ theory held that it must have been a microbe.

With that background, we arrive at London's 1854 cholera outbreak.

A particularly bad outbreak had hit the Soho district in London. If you are familiar with Soho, it is a rather upscale neighborhood today. However, in the mid-19th century, it was anything but.

It was a location of many animal processing facilities, including slaughterhouses, pens with animals, and rendering plants. There was tons of animal waste, decaying organic matter, and buildings with leaky cesspools.

Basically, the whole area was filthy, and most of this waste was flushed into a very underdeveloped sewer system.

Enter into the story, John Snow.

Snow was the bastard son of Eddard Stark and was raised at the Castle Winterfell. He had two brothers……

Wait, hold it, wrong Jon Snow.

John Snow was born to a poor family in York, England in 1813. However, he was very intelligent and managed to get an apprenticeship with a doctor in Newcastle-upon-Tyne at the age of 14.

At the time, being a physician was more of a trade than a science. You learned on the job and didn't necessarily have to attend school.

Snow encountered his first cholera epidemic in 1832 at a coal mining village called Killingworth.

He held several other positions and finally attended a medical school in London in 1836, earning his MD.

He had a very distinguished career becoming a member of the Royal College of Physicians, the Royal College of Surgeons, and was a founding member of the Epidemiological Society of London.

When the cholera outbreak occurred in 1854, it was the third such outbreak in recent memory, with outbreaks occurring in 1832 and 1849.

Snow didn't buy into the dominant theory of miasma and believed in the germ theory, even though there was still a lot that had to be figured out about germs.

He had written a paper in 1849 titled "*On the Mode of Communication of Cholera.*"

Snow did something very different in his approach to the outbreak. He approached it very analytically.

Rather than just treating this like every other cholera epidemic, he started collecting data. He methodically began to record all of the deaths and, most importantly, where they occurred.

He plotted all of the deaths on a map to see if he could determine if there was any pattern.

…and he discovered that there was. Several things, actually.

People in London received their water from several different companies. The companies which delivered water usually delivered it to everyone in a neighborhood regardless of income or social status.

Moreover, there were sharp divisions between who received water from which company.

Two homes on opposite sides of the street might have water provided by totally different companies.

Some companies would get their water from the Thames River, and other companies got their water from other sources. There were also very noticeable differences in how they filtered and purified their water. Some companies would deliver water with animal hairs and other visible impurities, and others would provide some filtration.

He looked at where the cholera deaths occurred and found that who you got your water from made a huge difference in who got sick and died.

He found that the Southwark and Vauxhall Waterworks and the Lambeth Waterworks Company both had very high rates of cholera amongst their customers. They both took their water from the Thames and didn't filter their water.

Other water companies, such as the New River Company and Chelsea Company, supplied better quality water and didn't have nearly as many cases of cholera.

What John Snow didn't know is that he had conducted one of the first double-blind studies. He was able to isolate the variable of water supply, and everything else was kept the same. The houses, the neighborhoods, and the people were pretty much the same across water companies.

On August 31, 1854, a particularly bad outbreak occurred in the blocks surrounding Broad Street, today called Broadwick Street.

Over a three-day period, 127 people died. By September 10th, 500 people had died.

Most of the residents fled the area to try to get away from the disease. Many of those who were sick were taken to Middlesex Hospital, where they were treated by one Florence Nightengale, who is considered to be the founder of modern nursing, and who will be the subject of a future episode.

Snow went into the neighborhood and started talking to people to try and get an idea of what might have been happening.

He also plotted the deaths on a map of the neighborhood and found that they all centered around a particular spot.

With the data he collected, and from talking to people who lived there, Snow figured out that the cause of the outbreak was a public water pump on the corner of Broad and Cambridge, now called Lexington Street.

Snow walked down to the pump and removed the handle so that no one could get water from it anymore.

Removing the handle basically stopped the outbreak in the neighborhood.

Snow later wrote in his own words:

On proceeding to the spot, I found that nearly all the deaths had taken place within a short distance of the [Broad Street] pump. There were only ten deaths in houses situated decidedly nearer to another street-pump. In five of these cases the families of the deceased persons informed me that they always sent to the pump in Broad Street, as they preferred the water to that of the pumps which were nearer. In three other cases, the deceased were children who went to school near the pump in Broad Street …

With regard to the deaths occurring in the locality belonging to the pump, there were 61 instances in which I was informed that the deceased persons used to drink the pump-water from Broad Street, either constantly or occasionally …

The result of the inquiry then was, that there had been no particular outbreak or prevalence of cholera in this part of London except among the persons who were in the habit of drinking the water of the above-mentioned pump-well.

It seemed like John Snow had built a slam dunk case for the germ theory of the spread of cholera by water.

However, after the outbreak subsided, the London Board of Health conducted an investigation into the cause of the epidemic.

Even after determining that the water companies had contaminated water, they concluded that the ultimate cause of the cholera outbreak was … miasma.

If the story sounds familiar, it might remind you of the previous episode on Ignaz Semmelweis, the 19th-century Hungarian doctor who discovered that washing your hands would decrease mortality rates when delivering babies.

Despite the very clear evidence that washing your hands worked, his conclusions were rejected by the medical establishment at the time.

Even though the work of John Snow wasn't immediately accepted, it eventually laid the foundation for the acceptance of the germ theory of disease just a few decades later.

Snow's work on tracking the causes of the cholera outbreak is considered to be the modern basis of the science of epidemiology. Today, whenever there is an outbreak of a disease, you'll find epidemiologists gathering data to try to find the source.

The findings of John Snow were later used for improvements in the water and waste management system. He also was responsible for the adoption of anesthesia in surgery.

Only four years after the outbreak in 1858, he died at the age of 45 of a stroke.

If you happen to be in London, you can visit the pump's actual location, which caused the Broad Street Cholera Outbreak. There is a replica of the pump located at the corner of Broadwick and Lexington Streets, and you'll know you are in the right place because it is right outside of a pub called the "John Snow."

There is also a John Snow society that occasionally meets at the John Snow pub.

While more people are probably familiar with Jon Snow as a character from Game of Thrones, more attention should probably be given to the real John Snow, who helped to pave the way for the creation of modern medicine.

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